Mobile Device Forensic Examination Report

EXTRACTION METHODS AND RESULTS

Case Identifier	DFIR-MF-2026
CASE TYPE	Personal Laboratory Examination
CLASSIFICATION	CONFIDENTIAL — LAW ENFORCEMENT SENSITIVE
Report Date	August 06, 2026
Examination Date(s)	August 05 to 06, 2026
PREPARED BY	Julian Derry
Case Reference	AMF-001

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1. Administrative & Case Details

1.1 Case Information

Item	Value
Case Identifier	DFIR-MF-2026
Requesting Agency / Client	Personal Laboratory Examination
Report Date	August 06, 2026
Examination Date(s)	August 05–06, 2026
Case Reference	AMF-001

1.2 Examiner Information

Item	Value
Examiner Name	Julian Derry
Title / Position	Digital Forensics Analyst
Organization	Independent / Personal Laboratory
Signature / Authorization	JD
Date Signed	August 06, 2026

1.3 Scope of Examination

This examination involved the forensic acquisition and analysis of a Samsung SM-A032F mobile device submitted for laboratory examination. Two acquisition methods, Advanced Logical and File System, were deliberately performed on the same device to evaluate and compare their respective data yields under the device's current security patch level. The examination was conducted using accepted digital forensic procedures intended to preserve data integrity and maintain evidential continuity within the limits of a personal laboratory environment.

1.4 Examination Objectives

• Evaluate and compare the data yield of Advanced Logical and File System extraction methods on a device with an updated security patch level. This evaluation was conducted on a device running Android 13 (Security Patch Level: 1 December 2025, Base Version: A032FXXS9CZA1) to assess how extraction method choice affects artifact recovery under current security conditions.

• Identify and preserve data stored on the device.

• Acquire forensic copies of accessible data.

• This evaluation focused on extraction method comparison at the data yield and artifact-count level, detailed analysis of individual artifact categories (communications, user activity, media, location, deleted data) was outside the scope of this exercise.

• Produce a documented forensic report suitable for technical review and evidential reference.

2. Evidence & Chain of Custody

2.1 Evidence Description

Item	Value
Evidence Item No.	AMF-001
Device Type	Mobile Phone
Manufacturer	Samsung
Model	SM-A032F A03 Core
Android Version	13
Security Patch Level	1 December 2025
Based Version	A032FXXS9CZA1
Color	Black
Physical Condition	Several cracks on phone screen

2.2 Device Identifiers

Item	Value
IMEI	35138xxxxxx5619
IMEI 2	35322xxxxxx5618
MEID	3513xxxxxx7561
ESN	Not present
Serial Number	R7xxxxxxR6A

Examiner Note

On this device, the IMEI and MEID appear together because the handset uses a hybrid identifier format in which the MEID is represented by the first 14 digits and the IMEI is represented by the full 15-digit value. ESN (Electronic Serial Number) is an older identifier format that has largely been replaced by MEID on modern devices.

2.3 SIM Card Details

Item	Value
SIM Slot	SIM 1 & SIM 2
ICCID (SIM 1)	Not obtained

Item	Value
ICCID (SIM 2)	8923 3020 5072 04XXXX
IMSI	6200xxxxxx58050 | 6200xxxxxx48328
Carrier	MTN | TELECEL

The SIM card was inserted at the time of extraction. Both SIM cards (SIM 1: TELECEL, SIM 2: MTN) were physically present in the device during extraction. However, neither tool recovered ICCID data for either SIM under either extraction method. Physical verification was later performed against the SIM tray, confirming SIM 2's ICCID as shown above (partially redacted for privacy). SIM 1's card did not display a standard-format ICCID on its label, only a batch/lot identifier, so its value remains listed as "Not obtained.

2.4 Chain of Custody Log

Date & Time	Action	From	To	Location	Signature
August 05, 2026 6:20 PM	Device photographed and documented	Julian Derry	Julian Derry, Digital Forensics Analyst	Accra, Ghana	JD
August 05, 2026 6:24 PM	Forensic extraction initiated	Julian Derry	Julian Derry, Digital Forensics Analyst	Accra, Ghana	JD
August 05, 2026 6:27 PM	Forensic extraction completed	Julian Derry	Julian Derry, Digital Forensics Analyst	Accra, Ghana	JD
August 05, 2026 6:30 PM	Device retained by examiner following examination	Julian Derry	Julian Derry, Digital Forensics Analyst	Accra, Ghana	JD

Examiner Note

This examination was conducted on a personally owned device within a controlled home laboratory environment. As a result, some chain-of-custody events commonly recorded in formal law-enforcement or corporate evidence handling procedures, such as seizure by an external party, evidence bag sealing, transport to evidence storage, and transfer to an investigating officer, did not occur and are therefore not recorded in this log.

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3. Intake State & Preservation

3.1 Device Condition on Receipt

Item	Value
Power State	On
Screen State	Unlocked
Battery Level	62%
SIM Present	Yes

Item	Value
External Damage	Multiple cracks on screen, scratches on body; small chip below power button

3.2 Isolation and Preservation Measures

• Device isolated from cellular networks.

• Wi-Fi and Bluetooth communications prevented.

• Airplane mode enabled.

• Device handled using forensic best practices to avoid data alteration.

3.3 Photographic Documentation

• Front view

• Rear view

• Left and right sides

• Top and bottom edges

• Screen display upon receipt

• Any visible damage

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4. Extraction & Acquisition Methodology

4.1 Acquisition Type

Acquisition Method	Performed
Logical Extraction	Yes (Advanced Logical)
File System Extraction	Yes
Physical Extraction	No

4.2 Hardware and Software Tools

Tool	Version	Purpose
Cellebrite UFED	7.71.0.1858	Data extraction
Cellebrite Physical Analyzer	8.1.0.12	Analysis of the UFED extraction file (.ufdx)

Examiner Note

The extraction package generated by Cellebrite UFED in .ufdx format was subsequently analyzed using Cellebrite Physical Analyzer, which supports examination of UFED extraction containers and associated artifact datasets.

4.3 Integrity Verification

Extraction File	Algorithm	Hash Value
Samsung GSM_SM-A032F A03	MD5 Hash	0BE3A4D8053951CB3E1F564C43EC1A80
	SHA-1 Hash	5B5840478187293F2B50A315CC8723844B900FFF

Hash values were calculated immediately after acquisition and verified prior to analysis.

4.4 Extraction Notes and Anomalies

Observation	Details
Connection Errors	None
Extraction Warnings	A backup option popup appeared during acquisition
Unsupported Artifacts	Instant messaging application data not acquired
Encryption Status	Encrypted
Acquisition Limitations	Device's security patch level limited privilege escalation during File System acquisition, resulting in a reduced dataset relative to Advanced Logical

Examiner Note

Advanced Logical: The Advanced Logical case was initially reviewed using the .ufdx file alone, which returned no media or file artifacts. A subsequent review incorporating the associated extraction archive (.zip) revealed the complete Device Data Files.

File System limitation: File system extraction was performed but limited by the device's security patch level, which restricted privilege escalation, resulting in a reduced dataset compared to the Advanced Logical extraction. File system extraction generally needs elevated privileges to reach protected storage, whereas Advanced Logical uses vendor-built routines that pull from structured app containers and database paths without needing that escalation, so it was not blocked in the same way.

Known vulnerabilities and public exploits were researched against this device's exact configuration, Android 13, Security Patch Level 1 December 2025, Base Version A032FXXS9CZA1. No unpatched public exploits were found to be actively viable against this build. This finding is consistent with the privilege-escalation limitations encountered during File System acquisition and explains why Physical/bootloader extraction was not pursued as part of this exercise.

The observations and limitations documented in this section apply specifically to the version of Cellebrite UFED used during this examination and should not be generalized to all versions of the software.

5. Technical Findings & Analysis

5.1 Communications

Call Logs

A total of 111 call records were acquired. The table below contains a representative sample of four records for reporting purposes.

Date/Time	Direction	Number / Contact	Duration
8/5/2026 1:12:00 PM (UTC+0)	Outgoing	116	00:05:36
8/2/2026 12:54:21 PM (UTC+0)	Outgoing	027xxxxxxx	
7/23/2026 11:31:12 AM (UTC+0)	Answered	020xxxxxxx	00:00:41
7/8/2026 9:48:46 AM (UTC+0)	Missed	059xxxxxxx	

SMS / MMS

A total of 374 SMS/MMS records were acquired. The table below contains a representative sample of three records for reporting purposes.

Date/Time	Sender	Recipient	Summary
8/5/2026 5:11:42 PM (UTC+0)	Telecel	05068xxxxx	Did you know, you can save more on every call…
8/5/2026 2:28:55 PM (UTC+0)	MTNFBB	05068xxxxx	Your request for broadband bundle purchase through MTN Mobile Money failed.
8/5/2026 9:32:03 AM (UTC+0)	505	05068xxxxx	Super news! More options for you!

Examiner Note

Dual-SIM Correlation: The device was provisioned with two active SIM profiles (IMSI 6200xxxxxx58050 on MTN and IMSI 6200xxxxxx48328 on TELECEL). Both carriers are represented in the acquired SMS dataset, MTN-originated messages (sender "MTNFBB") and TELECEL-originated messages (sender "Telecel") both appear among the extracted records, indicating that both SIM slots were actively in use rather than one being idle or a backup line. This is consistent with normal dual-SIM personal use rather than a single primary line with a secondary/unused SIM.

5.2 User Data

Contacts

A total of 1 contact record was acquired.

Name	Number	Email
Jude	02461xxxxx	

Examiner Note

The extraction contained only one contact record. During examination, it was confirmed that only one contact was intentionally stored in the device's internal contact storage. The extraction was permitted to access device contact data while SIM card contact storage was not accessible.

Calendar Entries

A total of 1 calendar event was acquired.

Date	Event	Notes
1/1/1970	Jude's birthday	

Examiner Confirmation

The examiner confirmed that only one calendar event had been manually added to the device prior to acquisition.

Additional user artifacts such as notes, reminders, and emails were not present.

5.3 Media and Metadata

Media Summary

Media Type	Advanced Logical	File System
Images	190	178
Audio Files	2	2
Videos	3	0
Documents	1	0
Archives	13	1
Text Files	85	1
Uncategorized	5	13

Findings

Media artifacts were recovered from both extractions, with the Advanced Logical extraction yielding a broader file set across nearly every category. A sample verification was performed. One video file from the Advanced Logical extraction and one audio file from the File System extraction were opened and confirmed to play successfully, indicating the recovered media is intact and not corrupted. A full content review of all recovered files, along with EXIF, geolocation, or timestamp metadata analysis, was not performed and was outside the scope of this examination.

Interpretation

Both extractions recovered media artifacts from the device, with Advanced Logical yielding a substantially broader file set than File System across nearly every category. The methodological reasons for this divergence, tied to the device's security patch level restricting privilege escalation during File System acquisition are discussed in Section 5.6.

5.4 Location Artifacts

Artifact Type	Details
GPS Records	0
Wi-Fi Networks	0
Cell Tower Associations	0
Significant Locations	0

Examiner Note

No location artifacts were recovered from either the Advanced Logical or File System extraction. Location artifacts are typically stored in system and app-level databases that neither acquisition method reliably captured on this device, consistent with the privilege-escalation limitation discussed in Section 4.4.

5.5 Data Recovery

Deleted records were examined using forensic recovery techniques, including database review and artifact analysis where applicable.

Artifact Type	Records Recovered	Recovery Method
SMS	374	Present in extraction, not identified as deleted
Contacts	1	Active device record
Photos	190 (Advanced Logical) / 178 (File System)	Active device records, not identified as deleted
Chat Messages	0	No recoverable deleted artifacts identified

Examiner Note

Both extractions recovered active photo files (190 via Advanced Logical, 178 via File System), but neither dataset flagged any of these, or any other artifact type, with a deletion status indicator. The photo counts represent active, currently-stored files, not recovered deleted media. As with SMS, no artifacts were conclusively identified as deleted records by the forensic tool during this examination.

The absence of recovered deleted artifacts in this examination should not be interpreted as confirmation that no deleted data exists on the device. Neither Advanced Logical nor the limited File System extraction parses unallocated space, where deleted records typically reside after removal from active databases. Recovering deleted SMS, media, or database records generally requires a full File System or Physical extraction with unrestricted privilege escalation, which this device's security patch level did not permit (see Section 4.4). Accordingly, the findings in this section reflect the limitations of the acquisition methods, not a confirmed absence of deleted data.

5.6 Cross-Method Comparison

Both Advanced Logical and File System extractions were performed on this device to evaluate whether the two methods produced consistent results, and to document the effect of the device's security patch level on extraction outcomes.

Device Data (Calendar, Call Log, Contacts, Device Info, Instant Messages) was identical across both extractions, 111 call records, 374 SMS/MMS records, 1 contact, 1 calendar entry, and 16 device info fields in each. This consistency indicates both methods reliably captured OS-level structured data regardless of acquisition type.

Device Data Files diverged significantly, as shown in Section 5.3. Advanced Logical recovered a substantially larger and more categorized file set (190 images, 85 text files, 13 archives, 3 videos, 1 document) compared to File System (178 images, 1 text file, 1 archive, 0 videos, 0 documents, with a higher Uncategorized count of 13 versus 5).

This divergence is attributed to the device's security patch level restricting the privilege escalation that File System extraction requires to reach protected app-specific storage. Advanced Logical extraction avoided this limitation by using vendor-built routines that query structured application containers and database paths directly, without needing elevated OS privileges. The higher Uncategorized count in the File System result is also consistent with a privilege-limited pull. Files were present but not fully parsed or categorized by the tool.

This result demonstrates that extraction method selection, and awareness of device patch level, has a direct and measurable effect on data yield, and that relying on a single extraction method may understate the data actually present on a device.

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6. Conclusion

Both Advanced Logical and File System extractions were performed on the device to preserve and analyze accessible data and to evaluate the consistency of results across acquisition methods. Communications records, user data, media artifacts, location artifacts, and potentially recoverable deleted data were examined within the scope of these acquisitions.

Device Data (calls, SMS, contacts, calendar, device info) was consistent across both methods. Device Data Files diverged, with Advanced Logical recovering a broader dataset than File System due to the device's security patch level restricting the privilege escalation required for a full File System pull (see Sections 4.4 and 5.6).

The examination was conducted using Cellebrite UFED version 7.71.0.1858 and Cellebrite Physical Analyzer version 8.1.0.12. The findings and limitations documented in this report are specific to the data accessible from the device at the time of examination and to the capabilities and limitations of that software version and the acquisition methods used. No conclusions have been drawn beyond the evidence available in the extracted datasets.

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7. Appendices

Appendix	Description
Appendix A	Automated Extraction Report
Appendix B	Hash Verification Output

Appendix	Description
Appendix C	Photographic Log
Appendix D	File System Inventory
Appendix E	Call Log Export
Appendix F	Message Database Export
Appendix G	Media File Index
Appendix H	Location Artifact Export
Appendix I	Screenshot Repository, organized in two folders by extraction method (Advanced Logical / File System), corresponding to Sections 4 and 5.
Appendix J	Photo of the SIM tray showing both Telecel and MTN SIM cards as physically installed in the device at time of extraction. ICCID and identifying numbers redacted for privacy. Included to document physical verification of SIM presence referenced in Findings.

End of Report